

Unit 3.4 The Chain rule: p 197

Note Title

2014/03/20

g differentiable at x and
 f differentiable at $g(x)$ then:
 $F = f \circ g$ or $F(x) = f(g(x))$ is
differentiable at x given by:

$$F'(x) = f'(g(x)) \cdot g'(x) \quad \text{OR}$$

If $y = f(u)$ and $u = g(x)$ then
 $y = f(g(x))$
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Examples:

① $y = \sqrt{x+1} = (\underline{x+1})^{1/2}$ $g(x) = x+1$
 $f(x) = x^{1/2}$
 $y' = \frac{1}{2}(\underline{x+1})^{-1/2} \cdot 1$ $f(g(x))$
 $= f(x+1)$
 $= \sqrt{x+1}$

② $y = \sin 2x$
 $y' = \cos \underline{2x} \cdot 2$

③ $y = \sin^3 x = (\sin x)^3$
 $y' = 3(\underline{\sin x})^2 \cos x$

$$\textcircled{4} \quad y = \sqrt{4-3t}$$

$$y' = \frac{1}{2}(4-3t)^{-1/2} \cdot -3$$

$$\textcircled{5} \quad y = e^{\sin 4x}$$

$$y' = e^{\sin 4x} \cdot \cos 4x \cdot 4$$

Combine the power rule with the chain rule (p200)

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} [f(x)]^n = n[f(x)]^{n-1} \cdot f'(x)$$

$$\textcircled{1} \quad \frac{d}{dx} \sqrt{x^2+1} = \frac{1}{2}(x^2+1)^{-1/2} \cdot 2x$$

$$\textcircled{2} \quad \frac{d}{dx} (6x^2+1)^{10} = 10(6x^2+1)^9 \cdot 12x$$

$$\textcircled{3} \quad y = \left(\frac{t-2}{t+1} \right)^3 = 3 \left(\frac{t-2}{t+1} \right)^2 \cdot \frac{1 \cdot (t+1) - (t-2) \cdot 1}{(t+1)^2}$$

We can do the same with:

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{f(x)} = e^{f(x)} \cdot f'(x)$$

$$\textcircled{1} \quad y = e^{\tan x} \quad y' = e^{\tan x} \cdot \sec^2 x$$

$$\textcircled{2} \quad y = e^{\frac{x}{\sin x}} \quad y' = e^{\frac{x}{\sin x}} \cdot \frac{1 \cdot \sin x - x \cos x}{(\sin x)^2}$$

$$\textcircled{3} \quad y = e^{\sqrt{\sin 2x}} \\ y' = e^{(\sqrt{\sin 2x})} \cdot \frac{1}{2} (\sin 2x)^{-1/2} \cdot \cos 2x \cdot 2$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \cos[f(x)] = -\sin[f(x)] \cdot f'(x)$$

Do the same with the other trigonometric functions.

$$\frac{d}{dx} \cos(10x+1) = -\sin(10x+1) \cdot 10$$

$$\frac{d}{dx} \sin^4(x^2+2) = 4[\sin^3(x^2+2)] \cdot \cos(x^2+2) \cdot 2x$$

$$x^3 \quad 3^x ?$$

$$\frac{d}{dx} a^x = a^x \ln a$$

P202

$$\frac{d}{dx} e^x = e^x$$

Proof (Remember $e^{\ln a} = a$ for $a > 0$)

$$\begin{aligned} \frac{d}{dx} (a)^x &= \frac{d}{dx} (e^{\ln a})^x \\ &= \frac{d}{dx} e^{(\ln a)x} \\ &= e^{(\ln a)x} \cdot \ln a \\ &= a^x \cdot \ln a \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} e^{f(x)} &= e^{f(x)} \cdot f'(x) \\ \frac{d}{dx} e^x &= e^x \cdot \ln e \\ &= e^x \end{aligned}$$

Example: ① $\frac{d}{dx} 3^x = 3^x \ln 3$ ② $\frac{d}{dx} \pi^x = \pi^x \ln \pi$

$$\frac{d}{dx} a^{f(x)} = a^{f(x)} \cdot \ln a \cdot f'(x)$$

$$1. \frac{d}{dx} (1.6)^{\sin x} = (1.6)^{\sin x} \times \ln 1.6 \times \cos x \quad (a^x)$$

$$2. \frac{d}{dx} \sqrt{1 + \pi^{\tan x}} = \frac{1}{2} (1 + \pi^{\tan x})^{-1/2} \cdot$$

$$\pi^{\tan x} \cdot \ln \pi \cdot \sec^2 x$$

$$3. \frac{d}{dx} \sqrt{x^2 + \pi^{\tan x}} = \frac{1}{2} (x^2 + \pi^{\tan x})^{-1/2} \cdot$$

$$(2x + \pi^{\tan x} \cdot \ln \pi \cdot \sec^2 x)$$

Worksheet:

1. $y = e^{2x}$

2. $f(x) = \sin e^{3x}$

3. $y = e^{\sin(x^4)}$

4. $f(t) = \frac{1}{t^4}$

5. $y = (t - t^3)^{10}$

6. $f(t) = \cos^4 3t$

7. $f(x) = 2^x$

8. $f(x) = 2^{\operatorname{cosec}(8x)}$

9. $f(x) = \cot(2^{1-5x})$

10. $f(t) = 7 \tan(3t) - 3^{\tan 7t}$

11. $f(x) = \operatorname{cosec} \frac{\pi}{3}$

12. $f(x) = \frac{\sqrt{x^3 + 1}}{7 - \tan 7x}$

Worksheet on differentiation - sum, product, quotient and chain rules 1

Find the derivatives of the following functions. DO NOT simplify your answer

1. $y = \left(\frac{1+x}{x^3} \right)^4$

2. $y = \frac{1}{(2-3x)^3}$

3. $f(t) = \cot(2^t)$

4. $f(t) = \tan(e^t) \cdot e^{\tan t}$

5. $f(v) = \sqrt{v^3 + \sec v}$

6. $f(s) = \frac{1}{1-3s}$

7. $f(x) = 2 \cos(e^{-4x})$

8. $f(x) = e^{\sin 4x} \cdot \cos(2x)$

9. $f(x) = 2^{\tan 3x}$

10. $f(x) = 5 \sec(3^x)$

11. $f(x) = \operatorname{cosec} \frac{\pi}{3}$

12. $f(x) = (1 + 2 \sin 5x)^3$

13. $f(x) = 3^{5x} + 5x^3 + 5^{3x}$

14. $f(x) = 5(1-2x)^{10}$

15. $f(x) = \frac{4 \sin x^4}{e^{4 \sin x}}$

16. $y = \sqrt{\cos(e^{3x+1})}$

17. $y = 2e^{\cot x} - 3 \cot(e^{-x})$

18. $f(x) = \sqrt[3]{\sec(4x)} \tan(x^4)$

19. $f(x) = \frac{\sqrt{x^3+1}}{7-\tan 3x}$

20. $y = \frac{2^{3x} \sin(4x)}{\cos(1-x) \operatorname{cosec}(e^{-x})}$